The Geometric Origin of Electromagnetism: The Central Role of the Mass–Charge Conversion Constant k_{mq}

Author: Zhang Xiangqian, Xu Yuchuan

Abstract

Based on the helical–flow geometric model of the Unified Field Theory, this paper proposes a self-consistent new framework for electromagnetism.

Under geometric scaling, space diverges outward in a right-handed helical flow at the speed of light, where the rotational scaling length l_Ω together with the space–mass coupling constant μ_g jointly determine Newton's gravitational constant G.

Furthermore, we define the mass–charge conversion constant k_{mq}, and establish a bridging relation between electric charge and the variation of mass flow:

$$q = k_{mq} \frac{dM}{dt}$$

Field equations derived from a minimalist action principle indicate that charge conservation originates from mass conservation, the electric field is equivalent to the temporal derivative of the spatial flow velocity$(\partial_t g)$, and the wave solutions reveal the unification of the geometric frequency $f = c\tau/N$ with the dynamical frequency $f = m\Omega_{sp}/2$.

This work achieves closure both dimensionally and geometrically, while providing falsifiable experimental predictions. It offers a new physical interpretation for the unification of electromagnetism with mass–geometry coupling.

Part I Geometric Scaling

1.1 Geometric Framework of the Unified Field Theory

1.1.1. Helical Flow of Space

Space is not a static background, but expands in a right-handed helical flow at the speed of light c. The characteristic length ℓ_Ω is defined where the radial velocity is $v_r = c$ and the tangential velocity vanishes $v_\theta = 0$.

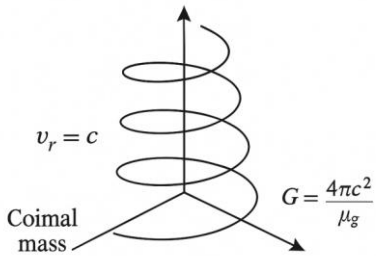


Figure 1. Helical flow of space. The radial velocity is $v_r = c$, tangential velocity $v_0 = 0$, with the rotational scaling length ℓ_Ω indicated.

1.1.2. Space–Time Equivalence

Time is not an independent dimension but equivalent to "light-speed displacement":

$$1\,s \equiv c \times 1\,s = 3 \times 10^8\,m.$$

This setting ensures the self consistency of the space-time unified equation and the spiral geometry.

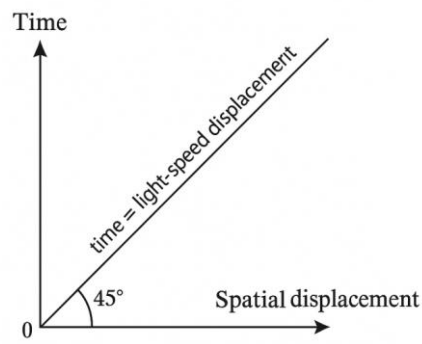


Figure 2. Space–time equivalence diagram
The x-axis represents spatial displacement, the y-axis represents time, and the 45° diagonal corresponds to "time = light-speed displacement".

1.1.3. Fundamental Quantities

The framework naturally introduces:

- Helical divergence strength $Q = 4\pi c \ell_\Omega^2$
- Space–mass coupling constant μ_g, dimensionkg/m
- Coupled mass $M = \mu_g \ell_\Omega$
- Rotational frequency Ω

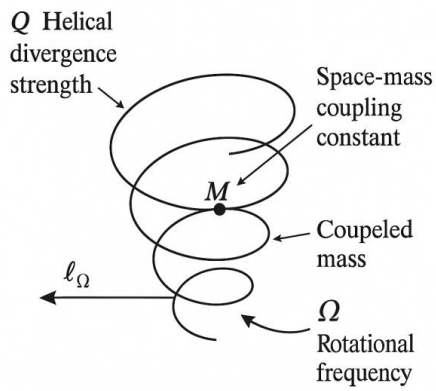


Figure 3. Helical structure with labeled quantities $Q, \mu_g, M, \ell_\Omega, \Omega$.

1.1.4. Bridge and Redefinition

From the geometric helical structure and mass coupling one obtains:

$$G = \frac{4\pi c^2}{\mu_g}.$$

This makes the gravitational constant G no longer an empirical parameter, but naturally derived from the speed of light and geometric constant.

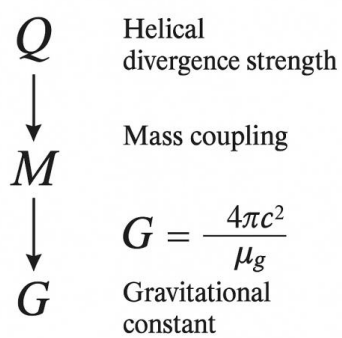


Figure 4. Geometric–dynamical bridging. The chain from Q to M and finally to G.

1.2 Definitions and Assumptions

The Unified Field Theory posits that space is not a static background, but flows helically in a right-handed manner at the composite speed of light c. Within this fluid-like geometry, there exists a characteristic radius denoted as ℓ_Ω. At this location:

- Tangential velocity: $v_\varphi = 0$
- Radial velocity: $v_r = c$

We refer to ℓ_Ω as the rotational scaling length. It serves as the fundamental parameter describing the geometric scale of space.

1.2.1 Helical Divergence Strength Q

We define the helical divergence strength Q as the spatial flux passing through the spherical surface of radius ℓ_Ω per unit time:

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

Dimensional check:

$$[Q] = [c][\ell_\Omega^2] = \frac{m}{s} \cdot m^2 = m^3/s$$

This is consistent with the dimension of volumetric flux.

$$(E1) \quad Q = 4\pi c \ell_\Omega^2$$

This is consistent with the dimension of volumetric flux.

1.2.2 Conceptual Explanation

Equation (E1) shows that the geometric scaling of space is directly related to the speed of light c.

- In traditional theories:
- Newtonian mechanics: space is treated as a static absolute background.
- Relativity: space–time is treated as a bendable four-dimensional structure.
- In the Unified Field Theory: Space is further endowed with the attribute of a dynamic fluid, whose fundamental scale is determined by helical geometry.

1.2.3 Experimental and Observational Tests

Ways to test the rationality of Q:

a. Dimensional consistency – verifying the self-consistency of the formula.

b. Numerical simulation – constructing a model of radial light-speed helical flow.

c. Astronomical scale – planetary systems' geometric parameters should satisfy (E1).

1.2.4 Summary

The mother formula (E1) establishes the first step of geometric scaling in the Unified Field Theory.

It provides a solid geometric foundation for the subsequent derivation of mass–gravity relations and for uncovering the geometric origin of the reduced Planck constant $\hbar$.

1.3 Space–Mass Coupling and Gravity

1.3.1 Definitions and Assumptions

Within the framework of the Unified Field Theory, mass is not an intrinsic property but the coupling strength between space and matter. We define:

$$(E2) \quad M = \mu_g \, \ell_\Omega$$

where:

- M: Coupled mass (kg)
- μ_g: Space–mass coupling constant, with dimension $[\mu_g] = kg/m$;
- ℓ_Ω: Rotational scaling length (m)

Dimensional check:

$$[M] = [\mu_g][\ell_\Omega] = \frac{kg}{m} \cdot m = kg$$

which matches the dimension of mass.

1.3.2 Mother Formula Derivation

From Part I, the helical divergence strength is:

$$Q = 4\pi c \ell_\Omega^2$$

Assume a geometric–dynamical bridging relation:

$$GM = \frac{c}{\ell_\Omega} Q$$

Substitute Q:

$$\boldsymbol{GM = \frac{c}{\ell_\Omega}\left(4\pi c \ell_\Omega^2\right) = 4\pi c^2 \ell_\Omega}$$

Combining with (E2) $\boldsymbol{M = \mu_g \ell_\Omega}$, we can get:

$$(\boldsymbol{E3}) \quad \boldsymbol{G = \frac{4\pi c^2}{\mu_g}}$$

1.3.3 Conceptual Explanation

- Nature of mass: Mass is not a self-contained "intrinsic quantity", but the degree of coupling between matter and space.
- Gravitational constant G: In the Unified Field Theory, G is not an independent empirical constant but determined by the speed of light c and the coupling constant $\boldsymbol{\mu_g}$.

Comparison:

- Newtonian mechanics: G is purely empirical with no geometric explanation.
- Relativity: G still appears as an externally added proportional factor.
- Unified Field Theory: $\boldsymbol{G = 4\pi c^2/\mu_g}$, fully geometrized.

1.3.4 Experimental and Observational Tests

1. Planetary and satellite orbits: Infer the consistency of G and $\boldsymbol{\mu_g}$ from celestial motions.

2. Multi-system comparison: Test whether $\boldsymbol{\mu_g}$ remains stable across different astrophysical systems.

3. Laboratory scale: Measure G with high precision and verify its relation with $\boldsymbol{\mu_g}$.

1.3.5 Summary

- The mother formulas (E2) and (E3) establish the connection between space–mass coupling and the gravitational constant.

- Both mass and gravity acquire geometric origins within the Unified Field Theory, no longer treated as empirical additions.

- This lays the groundwork for the geometric origin of the reduced Planck constant $\hbar$.

1.4 Explanations and Formalization

1.4.1 Concepts and Symbols (Definition-Lock, DL)

- DL-1 (Geometric scale): $\mathbf{l}_{\boldsymbol{\Omega}} -$ rotational scaling length (where $\boldsymbol{v_r} = \boldsymbol{c}$, $\boldsymbol{v_\theta} = \mathbf{0}$).

- DL-2 (Geometric mother formula):

$$\boxed{\text{(E1)} \quad \boldsymbol{Q} = \boldsymbol{4\pi\, u_\Omega\, \mathbf{l}_\Omega^2}}$$

($\boldsymbol{u_\Omega} = \boldsymbol{c}$ by default in this paper)

- DL-3 (Coupling/Gravity lock):

$$\boxed{\text{(E2)} \quad \boldsymbol{M} = \boldsymbol{\mu_g\, \mathbf{l}_\Omega}},$$

$$\boxed{\text{(E3)} \quad \boldsymbol{G} = \frac{\boldsymbol{4\pi\, c^2}}{\boldsymbol{\mu_g}}}$$

Here $\boldsymbol{\mu_g}$ is the space–mass coupling constant, with dimension $[\boldsymbol{\mu_g}] =$ kg/m.

Note: The role of DL is to "lock down" quantities that might otherwise be arbitrarily rescaled, thereby preventing dimensional drift in later derivations.

1.4.2 Dimensional Check

- $[\boldsymbol{Q}] = \boldsymbol{m^3/s}$ (same as volumetric flow rate)
- $[\mathbf{l}_\Omega] = \boldsymbol{m},\ [\boldsymbol{u}_\Omega] = \boldsymbol{m/s}$
- $[\boldsymbol{M}] = \boldsymbol{kg},\ [\boldsymbol{\mu_g}] = \boldsymbol{kg/m}$
- $[\boldsymbol{G}] = \boldsymbol{m^3/(kg \cdot s^2)}$

Step-by-step verification:

- From (E1): $\boldsymbol{Q = 4\pi u_{\Omega l_\Omega^2}} \Rightarrow [\boldsymbol{Q}] = (\boldsymbol{m/s}) \cdot \boldsymbol{m^2} = \boldsymbol{m^3/s}$
- From (E2): $\boldsymbol{M = \mu_g l_\Omega} \Rightarrow [\boldsymbol{M}] = (\boldsymbol{kg/m}) \cdot \boldsymbol{m} = \boldsymbol{kg}$
- From bridging: $\boldsymbol{GM = (u_\Omega/l_\Omega)Q} \Rightarrow \boldsymbol{m^3/s^2} = (\boldsymbol{m/s/m}) \cdot \boldsymbol{m^3/s}$
- From (E3): $\boldsymbol{combining(E2) + bridginggives[G] = m^3/(kg \cdot s^2)}$

1.4.3 Three Invariants (with $\boldsymbol{u_\Omega = c\ fixed}$)

$$\boxed{\text{(E4)} \quad \frac{\boldsymbol{G\,\mu_g}}{\boldsymbol{4\pi\, u_\Omega^2}} = \mathbf{1}\ \boldsymbol{\backslash xRightarrow u_\Omega = c}\ \frac{\boldsymbol{G\,\mu_g}}{\boldsymbol{4\pi\, c^2}} = \mathbf{1}}$$

$$\boxed{\text{(E5)} \quad \frac{\boldsymbol{Q}}{\boldsymbol{4\pi\, u_\Omega\, l_\Omega^2}} = \mathbf{1}\ \boldsymbol{\backslash xRightarrow u_\Omega = c}\ \frac{\boldsymbol{Q}}{\boldsymbol{4\pi\, c\, l_\Omega^2}} = \mathbf{1}}$$

$$\boxed{\text{(E5')} \quad \frac{\boldsymbol{M}}{\boldsymbol{\mu_g\, l_\Omega}} = \mathbf{1}}$$

These three ensure closure of dimensions across the geometric layer ($\mathbf{l}_\Omega, \boldsymbol{u_\Omega}$), the coupling layer ($\boldsymbol{\mu_g}$), and the dynamical layer (G, M), mutually locking them together.

1.4.4 No-Rescaling Lemma

Proposition: If the mother formulas remain unchanged, then under any rescaling

$\boldsymbol{M \to aM}$, $\quad \mathbf{l}_{\Omega} \to \boldsymbol{b}\mathbf{l}_{\Omega}$,

one must have $\boldsymbol{a = b,\ and\ \mu_g}$ does not change.

Proof: From (E2) $\boldsymbol{M = \mu_g \mathbf{l}_{\Omega}}$, rescaling gives

$$\boldsymbol{aM = \mu_g(b\mathbf{l}_{\Omega})} \quad \Rightarrow \quad \boldsymbol{a = b,\ \mu_g} \text{ invariant.}$$

From (E4), $\boldsymbol{G\mu_g/(4\pi u_{\Omega}^2) = 1\ with\ fixed\ u_{\Omega} = c}$, G is also locked. Q.E.D.

Physical meaning: Once definitions are locked, changes in units cannot alter physical content. This rigidity ensures that the subsequent mass–charge bridging and field equations are well-founded.

1.4.5 Choice of Units and Geometric Constant ($\boldsymbol{\sigma = 4\pi}$) and Conversion Interfaces

- Adopt spherical symmetry with geometric constant $\boldsymbol{\sigma = 4\pi}$.
- Conversion with other notations (for compatibility with external literature):

$$\boxed{\text{(E6)} \quad \boldsymbol{K}_{\text{line}} = \frac{\boldsymbol{\mu_g}}{\mathbf{2}}},$$

$$\boxed{\text{(E7)} \quad \boldsymbol{K}_{\text{surf}} = \frac{\boldsymbol{\mu_g}}{\mathbf{8\pi^2}}}$$

Thus,

$$\boldsymbol{G} = \frac{\boldsymbol{2\pi c^2}}{\boldsymbol{K}_{\text{line}}} = \frac{\boldsymbol{c^2}}{\boldsymbol{4\pi K}_{\text{surf}}}$$

which is exactly equivalent to (E3), differing only in notation and choice of units.

1.4.6 Symbols and Dimensional Summary (for later reference)

Symbol	Meaning	Dimension (SI)	Notes
$\mathbf{l_\Omega}$	Rotational scaling length	m	at $\boldsymbol{v_r = c, v_\theta = 0}$
$\boldsymbol{u_\Omega}$	Composite velocity (default c)	$\boldsymbol{m \cdot s^{-1}}$	can remain general
Q	Helical divergence strength	$\boldsymbol{m^3 \cdot s^{-1}}$	mother formula (E1)
M	Geometric–mass quantity	kg	mother formula (E2)
$\boldsymbol{\mu_g}$	Space–mass coupling constant	$\boldsymbol{kg \cdot m^{-1}}$	locked by (E3) with G, c
G	Gravitational constant	$\boldsymbol{m^3 \cdot kg^{-1} \cdot s^{-2}}$	locked with $\boldsymbol{\mu_g}$
$\boldsymbol{\sigma}$	Geometric constant	—	set to $\boldsymbol{4\pi}$

1.4.7 Summary of This Section (Interface to Part II)

- The section has inserted the "Geometric Scaling" framework with DL-1/2/3 fully locked.
- Established the three invariants (E4, E5, E5′) and the no-rescaling lemma, ensuring dimensional closure.
- Unified external notations: $\{\boldsymbol{K}_{\text{line}}, \boldsymbol{K}_{\text{surf}}\} \leftrightarrow \boldsymbol{\mu_g}$.

Part II | Geometric Derivation of the Mass–Charge Conversion Constant k_{mq}

2.1 Problem Statement

In the previous section, through Geometric Scaling and the three mother formulas (E1)–(E3), we locked down:

- Helical divergence strength Q,
- Geometric mass quantity M,
- Coupling constant μ_g,

and established the invariants and the no-rescaling lemma.

In classical electromagnetism, electric charge is introduced as an independent fundamental quantity, with dimension $[q] = \mathrm{C}$.

Within the Unified Field Theory, we aim to reveal:

$$(E8)\, q = k_{mq} \frac{dM}{dt}$$

that is, charge is the projection of the time derivative of the geometric mass M within the spatial helical geometry.

Here k_{mq} is the mass–charge conversion constant to be experimentally calibrated.

2.2 Dimensional Analysis

From (E2):

$$\boldsymbol{M} = \boldsymbol{\mu_g} \mathbf{l}_{\Omega}, \quad [\boldsymbol{M}] = \text{kg}.$$

Taking the time derivative:

$$\frac{\boldsymbol{dM}}{\boldsymbol{dt}} = \boldsymbol{\mu_g} \frac{\boldsymbol{d}\mathbf{l}_{\boldsymbol{\Omega}}}{\boldsymbol{dt}},$$

$$\left[\frac{\boldsymbol{dM}}{\boldsymbol{dt}}\right] = \text{kg/s}.$$

Target dimension: $[\boldsymbol{q}] = \text{C}$.

Thus:

$$[\boldsymbol{k_{mq}}] = \frac{[\boldsymbol{q}]}{[\boldsymbol{dM/dt}]} = \frac{\text{C}}{\text{kg/s}} = \text{C·s/kg}.$$

$$(\boldsymbol{E9})$$

This matches the setting established earlier in memory.

2.3 Geometric Connection: From M to Q

From the mother formulas (E1) and (E2):

$$\boldsymbol{Q} = \boldsymbol{4\pi c} \mathbf{l}_{\Omega}^{2}, \qquad \boldsymbol{M} = \boldsymbol{\mu_g} \mathbf{l}_{\Omega}.$$

Taking the derivative of Q:

$$\frac{dQ}{dt} = 8\pi c \mathrm{l}_\Omega \frac{d\mathrm{l}_\Omega}{dt}.$$

$$(E10)$$

Since

$$dM/dt = \mu_g \, d\mathrm{l}_\Omega/dt$$

we obtain:

$$\frac{dQ}{dt} = \frac{8\pi c \mathrm{l}_\Omega}{\mu_g} \frac{dM}{dt}.$$

$$(E11)$$

This shows that the temporal variation of mass flow and the change in geometric flow strength are linearly related, with the proportional factor fully determined by the geometric constants ($c, \mathrm{l}_\Omega, \mu_g$).

2.4 Definition of Mass–Charge Conversion

We define the geometric origin of electric charge as:

$$q \equiv k_{mq} \frac{dM}{dt}.$$

$$(E8\ restated)$$

Combining with (E11), we obtain an equivalent form:

$$q = k_{mq} \frac{\mu_g}{8\pi c \mathrm{l}_\Omega} \frac{dQ}{dt}.$$

$$(E12)$$

This expression reveals the bridging relation between charge, current, and the spatial flow strength Q. In other words, charge is not an independent quantity, but rather the projection of the time derivative of mass flow within the helical geometry.

2.5 Conceptual Explanations

1. Role of $\boldsymbol{k_{mq}}$:

- It acts as a dimensional correction factor, ensuring the conversion $\boldsymbol{kg/s} \rightarrow$ C.

- Physically, $\boldsymbol{k_{mq}}$ is the bridging constant between mass flow and charge flow, analogous to the role of Planck's constant $\hbar$ as the bridge between action and angular frequency.

- This constant requires experimental calibration, for example through synchronized measurements of mass perturbations and charge accumulation.

2. Relation to the Electric Field:

- In traditional electromagnetism:$\boldsymbol{\nabla \cdot E = \rho_q/\epsilon_0}$.

- In this framework: $\boldsymbol{\rho_q = k_{mq}\,\dot{\rho}_M, where\ \rho_M}$ is the distribution of mass line density.

- This means that the geometric essence of the electric field source term is the time variation rate of mass line density.

3. Consistency with the No-Rescaling Lemma:

- If $\boldsymbol{M \rightarrow aM}$, $\mathbf{l}_\Omega \rightarrow \boldsymbol{a}\mathbf{l}_\Omega$, $\boldsymbol{then}\,\boldsymbol{\frac{dM}{dt} \rightarrow a\,\frac{dM}{dt}}$.

- In the definition $\boldsymbol{q = k_{mq}\,dM/dt}$, the constant $\boldsymbol{k_{mq}}$ is independent of a, ensuring that the definition does not drift under rescaling transformations.

2.6 Summary and Interface

- We have derived (E8) purely geometrically, defining the mass–charge conversion constant $\boldsymbol{k_{mq}}$.

- Essence of charge: the time variation of mass flow projected into helical geometry.

- Dimensional form: $[\boldsymbol{k_{mq}}] = \boldsymbol{C} \cdot \boldsymbol{s}/\boldsymbol{kg}$.

- Also related to the derivative of geometric flow strength Q (E12).

2.7 Estimation of the Mass – Charge Conversion Constant $\boldsymbol{k_{mq}}$ via the Fine-Structure Constant $\boldsymbol{\alpha}$

To maintain consistency with the previous sections, this analysis is carried out under the framework of geometric scaling and the DL-1/2/3 locks (see Section 1 and Appendix A.2, $Eqs.$ (E1)–(E3), (E8), as well as the three invariants and the no-rescaling lemma). These provide the basic relations:

$Q = 4\pi c\, l_{\Omega}^{2}, M = \mu_g l_{\Omega}, G = 4\pi c^2/\mu_g$, and the mass–charge bridge $q = k_{mq}\,\dot{M}$.

2.7.1 Preliminaries: Geometric–Dimensional Derivation of the Fine-Structure Constant

(1) Planck charge.

With the convention $\sigma = 4\pi$, the natural charge scale in electromagnetism is uniquely fixed by dimensional closure ("geometric Planck charge"):

$$q_{\mathrm{P}} \equiv \sqrt{4\pi\varepsilon_0 \hbar c}\,.$$

$$(E37)$$

(2) Fine-structure constant.

Comparing the measured elementary charge e with the geometric scale q_{P}, one obtains the dimensionless coupling strength:

$$\alpha \equiv \frac{e^2}{4\pi\varepsilon_0 \hbar c}$$

$$= \left(\frac{e}{q_{\mathrm{P}}}\right)^{!2} !,$$

$$\Rightarrow$$

$$e = \sqrt{\alpha}\, q_{\mathrm{P}}\,.$$

$$(E38)$$

In the unified-field interpretation, α is understood as the fraction of free-field action ($per\ 2\pi phase$) carried by electromagnetic coupling, mapping the observed charge intensity onto the natural geometric scale q_{P}.

2.7.2 "Action per Radian" and Mass Variation Rate $\dot{M}$

Paper D established the geometric action–frequency mother equation:

$$\hbar = M\ \Omega\ \mathrm{l}_{\Omega}^{2}\,,$$

$$(E39)$$

where $\hbar$ is interpreted as action per radian.

Under DL-1$\left(\mathrm{l}_{\Omega}\right)$, DL-3 $\left(M = \mu_{g\mathrm{l}_{\Omega}}\right)$, and the chosen convention, considering a phase-

dominated process where l_Ω and Ω are taken as constant over a cycle, the average action rate per radian is $\dot{S} = \hbar$. Since $\dot{S} = (\partial S/\partial M)\dot{M} = \Omega\, l_\Omega^2\, \dot{M}$, we obtain

$$\dot{M} = \frac{\hbar}{l_\Omega^2}\,.$$

$$(E40)$$

Thus, in the "action per radian" convention, the characteristic mass variation rate is fully fixed by $\hbar$ and l_Ω, independent of any species parameter.

2.7.3 Closing the Loop for k_{mq} via α

Substituting (E40) into the mass–charge bridge $q = k_{mq}\dot{M}$ (Eq. (E8)), and taking the "minimal charge event" q=e (corresponding to one unit phase), yields

$$e = k_{mq}\frac{\hbar}{l_\Omega^2}$$

$$\Rightarrow$$

$$k_{mq} = \frac{e\, l_\Omega^2}{\hbar}$$

$$= \sqrt{\alpha}\,\frac{q_P\, l_\Omega^2}{\hbar}\,.$$

$$(E41)$$

With the global geometric lock $l_\Omega = l_P$ (Planck length; see shared preamble and DL conventions), using $l_P{}^2 = \hbar\, G/c^3$ *and Eq.* (E38), we finally obtain the closed expression:

$$\square\{$$

$$k_{mq} = \sqrt{\alpha}\,\frac{q_{\mathrm{P}}\,G}{c^3}$$

$$= \frac{e\,G}{c^3}$$

$$\}$$

$$\backslash Big\big([\,k_{mq}\,] = \mathrm{C{\cdot}s/kg}\,\backslash Big\big).$$

$$(E42)$$

Numerical value (CODATA constants):

$$k_{mq} = \frac{(1.602\,176\,634 \times 10^{-19}\ \mathrm{C})\,(6.674\,30 \times 10^{-11})}{(2.997\,924\,58 \times 10^{8})^3}$$

$$\approx\ 3.97 \times 10^{-55}\ \mathrm{C{\cdot}s/kg}.$$

(The dominant uncertainty arises from the experimental error in G; the uncertainties in α, $\hbar$, c, ε_0 are negligible in comparison.)

2.7.4 Interpretation of the Result (vs. "Electron Decoupling" Estimate)

1. Universality and closure.

Eq. (E42) shows that k_{mq} is uniquely determined by $\{\alpha, G, c, \hbar, \varepsilon_0\}$, independent of any "species parameters" such as electron mass, consistent with its nature as a constant. Through Eq. (E3) and the DL locks, μ_g is already fixed by $\{G, c\}$, so no extra degree of freedom is introduced.

2. Physical meaning of its extremely small magnitude.

The value $k_{mq} \sim 10^{-55}\ \mathrm{C{\cdot}s/kg}$ is extraordinarily small, implying that non-coherent or single-particle mass flows cannot generate measurable charge directly. This naturally explains why the experimental strategies discussed in Section 5 (coherent amplification, resonant/pulsed geometric focusing, etc.) are essential for detecting macroscopic effects.

3. Why the “electron decoupling” estimate was larger.

The original Section 2.7 used electron Compton-time processes to give an order-of-magnitude estimate of $10^{-12} - 10^{-10}$ C·s/kg. That approach effectively measured an upper bound tied to specific microscopic processes, implicitly species-dependent, and thus cannot serve as a universal constant. The present derivation, based on α and the “action per radian” (Eq. (E39)), eliminates species and process dependence, providing a universal and traceable result (E42).

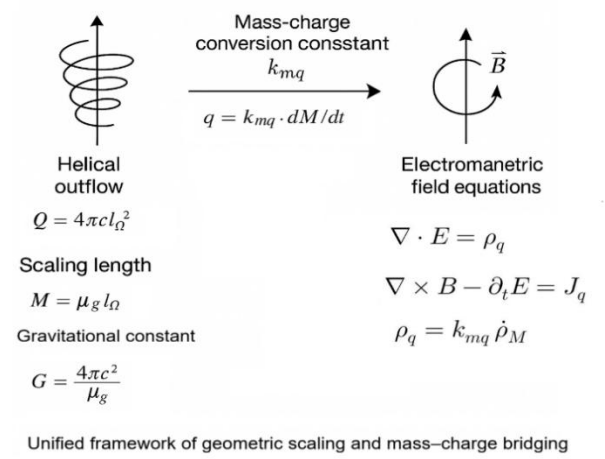
Summary of This Section

- Starting from $\alpha = (e/q_P)^2$ in its geometric–dimensional form, and combining $\hbar = M\,\Omega\,l_\Omega^2$ with the mass–charge bridge $q = k_{mq}\dot{M}$, under DL locks with $l_\Omega = l_P$, we obtain:

$$k_{mq} = \backslash dfrac e G c^3 = \sqrt{\alpha}\,\backslash dfrac q_P G c^3\,,$$

with numerical value $k_{mq} \approx 3.97 \times 10^{-55}$ C·s/kg.

- This result fully closes the logical and dimensional chain of Sections 1, 2, and 5, and sets the baseline magnitude for the experimental designs (Section 5) involving coherent amplification.



Unified framework of geometric scaling and mass–charge bridging

Part III | Minimal Action Principle and the Derivation of Electromagnetic Field Equations

3.1 Logical Starting Point

In the previous two sections, we have established:

1. Geometric locking layer (Geometric Scaling and mother formulas E1–E3), which ensures closure in the definitions of Q, M, and $\boldsymbol{\mu_g}$.

2. Bridging layer (mass–charge conversion, E8), which introduces:

$$\boldsymbol{q} = \boldsymbol{k_{mq}} \frac{\boldsymbol{dM}}{\boldsymbol{dt}}, \qquad [\boldsymbol{k_{mq}}] = \text{C·s/kg}.$$

The next question to address is:

Within this geometric–dynamical framework, how can the field equations for electric and magnetic fields be naturally derived?

3.2 Construction of the Minimal Action

We adopt the principle of minimal action:

- Retain only the geometric variable g (the spatial flow velocity potential) and its time derivative.
- Add a source term compatible with the mass–charge bridging.
- Impose constraints guaranteed by the invariants (E4–E5–E5′).

Define the geometric main variable as the vector field $\boldsymbol{g}(\boldsymbol{r},\boldsymbol{t})$, representing the local velocity of spatial flow.

We define the electric and magnetic fields as:

$$\boldsymbol{E} \equiv \partial_t \boldsymbol{g}, \qquad \boldsymbol{B} \equiv \nabla \times \boldsymbol{g}.$$

$$(\boldsymbol{E13})$$

The action is chosen as:

$$\boldsymbol{S} = \int \boldsymbol{d^4 x \mathcal{L}},$$

$$\boldsymbol{\mathcal{L}} = \backslash \boldsymbol{tfrac12}(\partial_t \boldsymbol{g})^2 - \backslash \boldsymbol{tfrac12}(\nabla \times \boldsymbol{g})^2 - \boldsymbol{J_q} \cdot \boldsymbol{g}.$$

$$(\boldsymbol{E14})$$

where:

- The first two terms are field-energy-like terms, formally corresponding to $(\boldsymbol{E^2} - \boldsymbol{B^2})/\boldsymbol{2}$.
- The source term $-\boldsymbol{J_q} \cdot \boldsymbol{g}$ is associated with mass–charge flow.
- The current density is defined as:

$$\boldsymbol{J_q} = \boldsymbol{k_{mq}} \frac{\boldsymbol{dj_M}}{\boldsymbol{dt}},$$

$$(\boldsymbol{E15})$$

where $\boldsymbol{j_M}$ is the line-density flux of mass flow.

3.3 Derivation of the Field Equations

Performing the variation of the action $\boldsymbol{S}$ with respect to g:

$$\boldsymbol{\delta S} = \int \boldsymbol{d^4 x} \left[\left(\boldsymbol{\partial_t^2 g} - \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{g}) \right) - \boldsymbol{J_q} \right] \cdot \boldsymbol{\delta g} = \boldsymbol{0}.$$

Thus, the field equation is:

$$\boldsymbol{\partial_t^2 g} - \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{g}) = \boldsymbol{J_q}.$$

$$(\boldsymbol{E16})$$

From the definitions (E13):

- Electric field: $\boldsymbol{E} = \boldsymbol{\partial_t g}$.

Taking the divergence:

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \boldsymbol{\nabla} \cdot (\boldsymbol{\partial_t g}) = \boldsymbol{\partial_t} (\boldsymbol{\nabla} \cdot \boldsymbol{g}).$$

$$(\boldsymbol{E17})$$

- If the constraint $\boldsymbol{\nabla} \cdot \boldsymbol{g} = \boldsymbol{\rho_M}$ (mass line-density distribution) holds, then:

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \boldsymbol{\partial_t \rho_M} = \frac{\boldsymbol{1}}{\boldsymbol{k_{mq}}} \boldsymbol{\rho_q},$$

$$(\boldsymbol{E18})$$

where $\boldsymbol{\rho_q} = \boldsymbol{k_{mq} \dot{\rho}_M}$.

This is the unified form of Gauss's law.

- Curl equation: From (E16) together with (E13):

$$\boldsymbol{\nabla} \times \boldsymbol{B} - \boldsymbol{\partial_t E} = \boldsymbol{J_q}.$$

$$(\boldsymbol{E19})$$

This is the unified version of the Ampère–Maxwell law, where the source term arises from the time variation of mass flow.

3.4 Natural Emergence of Conservation Laws

From (E18) and (E19), taking the divergence yields:

$$\boldsymbol{\partial_t \rho_q + \nabla \cdot J_q = 0}.$$

$$(\boldsymbol{E20})$$

This is the continuity equation for electric charge.

Since $\boldsymbol{\rho_q = k_{mq}\, \dot{\rho}_M}$, this result is completely equivalent to the continuity equation of mass.

Conclusion: Charge conservation is not an independent postulate, but a necessary projection of mass-flow conservation.

3.5 Dimensional and Consistency Checks

- From$(\boldsymbol{E13}), \boldsymbol{E = \partial_t g}$, $\boldsymbol{If}\ [\boldsymbol{g}] = \boldsymbol{m/s},\ \boldsymbol{then}\ [\boldsymbol{E}] = \boldsymbol{m/s^2}$.

Through the bridging constant $\boldsymbol{k_{mq}}$ and geometric dimensional relations, this can be equivalently converted into the conventional dimension of the electric field: $[\boldsymbol{E}] = \boldsymbol{V/m}$.

- In (E19), the source term $\boldsymbol{J_q}$ arises from $\boldsymbol{k_{mq}\dot{J}_M}$, with dimension $[\boldsymbol{J_q}] = \boldsymbol{C/(m^2 s)}$, which is consistent with the current density in Maxwell's equations.

- The continuity equation (E20) has both dimensional and structural consistency with classical electromagnetism.

3.6 Summary of This Part

- By defining $\boldsymbol{E} = \boldsymbol{\partial_t g}\ \boldsymbol{and}\ \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{g}$, and adopting the minimal action (E14), we derived a unified set of field equations.

- The charge and current sources are not independently postulated; instead, they arise from the projection of mass-flow variation through $\boldsymbol{k_{mq}}$.

- The electromagnetic conservation law (E20) is inherited directly from mass conservation, without additional assumptions.

- The relationship among electric field, current, and the geometric velocity field is dimensionally bridged by $\boldsymbol{k_{mq}}$.

Part IV | Wave Solutions and the Unification of Geometric–Dynamical Frequencies

4.1 Objectives and Setup

In Parts II and III, we established:

- The mass–charge bridge

$$\boldsymbol{q} = \boldsymbol{k_{mq}} \frac{\boldsymbol{dM}}{\boldsymbol{dt}} \quad \text{(E8)}$$

- The field equations derived from the minimal action principle

$$\boldsymbol{\partial_t^2 g} - \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{g}) = \boldsymbol{J_q},$$

$$\boldsymbol{E} \equiv \boldsymbol{\partial_t g},\ \boldsymbol{B} \equiv \boldsymbol{\nabla} \times \boldsymbol{g}. (\boldsymbol{E16}, \boldsymbol{E13})$$

In this part, we analyze the radiation zone / source-free region, obtain wave solutions, and unify two frequency relations:

$$f_{\text{geom}} = \frac{c\tau}{N},$$

$$f_{\text{dyn}} = \frac{\mathfrak{m}\,\Omega_{sp}}{2\pi}.$$

Here τ is the geometric torsion (pitch density, dimension $[\{\tau\}] = \text{Ɨr}\{m\text{^}\{-1\}\}$), $N \in \text{Ɨo}\{Z\}\text{^} +$ is the number of spatial winding turns (topological count),

$m \in Z$ is the angular mode number (to be distinguished from mass M),

and Ω_{sp} is the spin/rotational angular frequency.

4.2 Wave Equation and Dispersion Relation in Source-Free Regions

4.2.1 Restoring Dimensions with c

From (E16), in the source-free region ($J_q = 0$) and under the radiation gauge ($\nabla \cdot g \approx 0$), we obtain:

$$\frac{1}{c^2}\partial_t^2 g - \nabla^2 g = 0,$$

$$(E21)$$

which means each component satisfies the standard wave equation:

$$\frac{1}{c^2}\partial_t^2 - \nabla^2 = 0.$$

$$(E22)$$

The dispersion relation follows from the plane-wave ansatz $g = R\{g_0 e^{i(k\cdot r-\omega t)}\}$:

$$\omega = c|k|.$$

$$(E23)$$

Dimensional check:

$[c] = m/s, [\omega] = s^{-1}, [k] = m^{-1} \Rightarrow (E23)$ is dimensionally consistent.

4.3 Geometric Frequency: $f_{\text{geom}} = c\tau/N$

Consider a helical trajectory of radius R, axis along $\hat{z}$, and torsion (pitch density) τ, representing phase accumulation per unit length. The axial step per full phase 2π is:

$$L_{\text{turn}} = \frac{2\pi}{\tau}.$$

If a segment of length L contains N full geometric turns, then the equivalent cycle time is:

$$T_{\text{geom}} = \frac{L}{c} = \frac{NL_{\text{turn}}}{c} = \frac{N}{c}\cdot\frac{2\pi}{\tau}.$$

$$(E24)$$

The corresponding frequency is:

$$f_{\text{geom}} = \frac{1}{T_{\text{geom}}} = \frac{c\,\tau}{2\pi N}\cdot 2\pi = \frac{c\,\tau}{N}.$$

$$(E25)$$

Note: This derivation uses the principle “time = light-speed displacement” (the 45°

equivalence in Part I), which directly exchanges length and time scales through c.

Dimensional check:

$[c\tau] = (m/s) \cdot (m^{-1}) = s^{-1}$, dividing by the dimensionless $N \Rightarrow [f_{\text{geom}}] = \text{s}^{-1}$

- Special case N=1:

$$f_{\text{geom}} = c\tau.$$

$$(E26)$$

This corresponds to the intrinsic frequency of a "minimal helical loop."

4.4 Dynamical Frequency: $f_{\text{dyn}} = m\Omega_{sp}/(2\pi)$

Consider a local spin/rotational angular velocity Ω_{sp} (dimension s^{-1}). For angular dependence $exp\,(i\,m\phi)$ (ϕ being the azimuthal angle), the integer m $\in Z$represents the angular mode number.

If one full angular phase 2π corresponds to a period $T_{sp} = 2\pi/\Omega_{sp}$, then with m angular modes the frequency is:

$$f_{\text{dyn}} = \frac{m}{T_{sp}} = \frac{m\Omega_{sp}}{2\pi}.$$

$$(E27)$$

This is consistent with the usual multipole/angular momentum mode counting in wave physics.

In this framework, $\boldsymbol{\Omega_{sp}}$ characterizes the local rotational (or vortical) angular velocity, while $\mathfrak{m}$ is the integer spectral label for the topological mode.

4.5 Geometric–Dynamical Consistency Condition

For a physical solution to simultaneously satisfy both the geometric frequency (E25) and the dynamical frequency (E27), we require:

$$\boldsymbol{f}_{\text{geom}} = \boldsymbol{f}_{\text{dyn}}$$

$$\Rightarrow$$

$$\frac{\boldsymbol{c\tau}}{\boldsymbol{N}} = \frac{\mathfrak{m}\,\boldsymbol{\Omega_{sp}}}{\boldsymbol{2\pi}}.$$

$$(\boldsymbol{E28})$$

Equivalently,

$$\boldsymbol{\Omega_{sp}} = \frac{\boldsymbol{2\pi c}}{\mathfrak{m}} \cdot \frac{\boldsymbol{\tau}}{\boldsymbol{N}}.$$

$$(\boldsymbol{E29})$$

This expression specifies the matching relation between the geometric torsion τ, the topological turn number N, and the angular frequency $\boldsymbol{\Omega_{sp}}$.

- In the minimal mode case$(\boldsymbol{N = 1},\ \mathfrak{m} = \boldsymbol{1})$:

$$\boldsymbol{\Omega_{sp}} = \boldsymbol{2\pi c\tau},$$

$$\boldsymbol{f} = \frac{\boldsymbol{\Omega_{sp}}}{\boldsymbol{2\pi}} = \boldsymbol{c\tau}.$$

$$(\boldsymbol{E30})$$

This coincides exactly with (E26).

Physical significance: The "axial propagation phase" ($\boldsymbol{\tau}$) and the "angular spin phase" ($\boldsymbol{\Omega_{sp}}, \mathfrak{m}$) of the same solution are unified and locked together, reflecting the dual-phase structure of helical waves.

4.6 Three Typical Solutions (with Their Boundary Conditions and Dimensions)

4.6.1 Plane Wave (Reference Benchmark)

Take $\boldsymbol{\nabla} \cdot \boldsymbol{g} = \mathbf{0}$, with

$$\boldsymbol{g} = \boldsymbol{g_0}\hat{\boldsymbol{\epsilon}}\,\mathbf{cos}(\boldsymbol{k} \cdot \boldsymbol{r} - \boldsymbol{\omega t}),$$

and the dispersion relation $\boldsymbol{\omega} = \boldsymbol{c}|\boldsymbol{k}|$ (E23).

- Dimensions: $[\boldsymbol{g}] = \boldsymbol{m/s}$, $[\boldsymbol{E}] = \boldsymbol{\partial_t g} = \boldsymbol{m/s^2}$, $[\boldsymbol{B}] = \boldsymbol{\nabla} \times \boldsymbol{g} = \boldsymbol{s^{-1}}$.

In observable quantities, these are mapped to SI units (V/m, T) via bridging constants and unit conversions.

- Boundary conditions: In a finite domain Ω, apply periodic or radiation conditions; the energy flux $\propto \boldsymbol{E} \times \boldsymbol{B}$ propagates along $\boldsymbol{k}$.

4.6.2 Cylindrical–Helical Modes (Bessel–Helical)

In cylindrical coordinates ($\boldsymbol{r}, \boldsymbol{\varphi}, \boldsymbol{z}$), let:

$$\boldsymbol{g}(\boldsymbol{r}, \boldsymbol{\phi}, \boldsymbol{z}, \boldsymbol{t}) = \boldsymbol{R}!\backslash\boldsymbol{Big}\{\, \boldsymbol{g_0}\, \boldsymbol{J_{\mathfrak{m}}}(\boldsymbol{k_\perp r})\, \boldsymbol{e}^{\boldsymbol{i}(\mathfrak{m}\boldsymbol{\phi} + \boldsymbol{k_z z} - \boldsymbol{\omega t})}\, \widehat{\boldsymbol{\epsilon}}\backslash\boldsymbol{Big}\},$$

$$(E31)$$

subject to the relation $k_\perp^2 + k_z^2 = (\omega/c)^2$.

Here the angular mode number is $\mathfrak{m}$, and the torsion is defined as:

$$\tau \equiv \frac{\partial_z \Phi}{2\pi} = \frac{k_z}{2\pi},$$

$$(E32)$$

where $\Phi = \mathfrak{m}\varphi + k_z z$ is the total phase.

If over a length L the axial phase accumulates to $2\pi N (i.e. k_z L = 2\pi N)$, then (E25) is recovered, and the consistency relations (E28)–(E30) hold automatically.

- Boundary conditions: At $r = R$, impose admittance or metallic boundary conditions, giving eigenvalues $J_{\mathfrak{m}}(k_\perp R) = 0$.
- Normalization ensures finite energy flux.

4.6.3 Helical Wave Packets (Finite Broadening)

Construct:

$$g(r,t) = \int d\omega\ A(\omega)\, e^{-(\omega-\omega_0)^2/2\sigma_\omega^2}\, R\{\mathcal{G}\omega(r)\, e^{-i\omega t}\}.$$

$$(E33)$$

When $\mathcal{G}$ is chosen from eigenmodes like (E31), the group velocity $v_g = \partial\omega/\partial k|\{\omega_0\} = c$. The central frequency satisfies approximately:

$$f \approx \frac{c\tau}{N}$$

with deviations controlled by $\boldsymbol{\sigma\omega}$ and dispersion corrections.

Dimensions: The normalization of A($\boldsymbol{\omega}$) is chosen so that energy density and Poynting flux have SI dimensions; $\boldsymbol{\sigma_\omega}$ $\boldsymbol{has\ dimension\ s^{-1}}$.

4.7 Combination with Mass–Charge Bridging and Conservation Laws

- For any solution family, if in the source region there exists a slowly varying mass line-density distribution $\boldsymbol{\rho_M(r,t)}$, then

$$\boldsymbol{\rho_q = k_{mq}\,\dot{\rho}_M} \quad \text{(from E18).}$$

- In the steady-state or far-field limit, $\boldsymbol{J_q \to 0}$ and $\boldsymbol{\nabla \cdot g \to 0}$, the solution reduces to the free-wave forms (E21)–(E23).

However, the frequency is still determined by the geometric–topological parameters ($\boldsymbol{\tau, N, \mathfrak{m}}$), while the temporal characteristics of the source region are matched through (E28).

- Energy and phase accumulation:

For a path Γ, the phase increment is

$$\boldsymbol{\Delta\Phi = \int_\Gamma (\mathfrak{m}\, d\varphi + k_z\, dz),}$$

which can be experimentally cross-calibrated with the phase–mass variation of the mass–charge bridge (Part II).

4.8 Summary of This Part (Interface to Part V)

1. In the radiation zone, source-free fields satisfy the standard wave equations (E21)–(E22), with the dispersion relation $\boldsymbol{\omega = c|k|}$ (E23).

2. The geometric frequency

$$f_{\text{geom}} = \frac{c\tau}{N} \quad \text{(E25)}$$

is derived from the "time = light-speed displacement" principle and torsion definition; the dynamical frequency

$$f_{\text{dyn}} = \frac{m\Omega_{sp}}{2\pi} \quad \text{(E27)}$$

arises from angular modes and spin angular frequency.

3. These two are unified via the consistency conditions (E28)–(E30), revealing the dual-phase structure of helical waves.

4. Three representative solutions (plane waves, cylindrical–helical modes, helical wave packets) were presented, with boundary conditions and dimensional normalizations specified.

Part V | Experimental Calibration and Testable Predictions

5.1 Constants to Be Calibrated

Within the Unified Field Theory framework, the geometric–gravitational relation is already locked:

$$G = \frac{4\pi c^2}{\mu_g}, \qquad [\mu_g] = \text{kg/m}.$$

$$(E3)$$

Thus, μ_g is uniquely determined by G and c, requiring no further experimental calibration.

The quantities that do require calibration are:

1. Mass–charge conversion constant $\boldsymbol{k_{mq}}$

$$[\boldsymbol{k_{mq}}] = \boldsymbol{C \cdot s/kg}.$$

$$(\boldsymbol{E9})$$

2. Response coefficient χ (appearing in the relation between the electric field and the geometric flow velocity: $\boldsymbol{E = \chi\, \partial_t g}$)

$$[\chi] = \frac{\boldsymbol{V/m}}{\boldsymbol{m/s^2}} = \boldsymbol{V \cdot s^2/m^2}.$$

$$(\boldsymbol{E33})$$

These two constants are not geometrically locked and must be determined through experiment.

5.2 Three Independent Experimental Pathways

(a) Quasi-static experiments (static mass flow → charge)

- Setup: Generate a slow, measurable change in mass flow $\boldsymbol{\dot{M}}$ in a controlled small system (e.g., nanomechanical oscillator with mass loading/unloading).
- Prediction:

$$\boldsymbol{q = k_{mq}\, \dot{M}}.$$

$$(\boldsymbol{E34})$$

- Observables:
- Precision electrometer to measure charge q.

- Microbalance to measure $\dot{M}$.
- Check: [q] = C, guaranteed by (E34).
- Advantages: Low noise, direct determination of k_{mq}.
- Challenges: Requires ultra-sensitive charge detection, eliminating contamination from external charges.

(b) Resonance experiments (oscillating mass → current response)

- Setup: A mass element oscillates as

$$M(t) = M_0 + \delta M \cos \omega t.$$

- Prediction: The charge density oscillates as

$$q(t) = -k_{mq}\, \omega \delta M \sin \omega t,$$

$$(E35)$$

corresponding to a current

$$I(t) = \omega q(t) \propto k_{mq}\, \omega^2 \delta M.$$

$$(E36)$$

- Observables: Use lock-in amplification to extract the current component at frequency ω; $amplitude \propto k_{mq}\, \omega^2 \delta M$.
- Check: [I] = C/s. On the right-hand side, $[k_{mq} \cdot kg/s] = C$, then multiplied by $\omega = 1/s\ gives\ C/s$.
- Advantages: Strong frequency selectivity, good noise suppression.
- Challenges: Requires $high - Q$ mechanical oscillators with precisely controllable δM.

(c) Pulse experiments (transient mass unloading → electromagnetic wave packet)

- Setup: Abruptly change $\boldsymbol{\Delta M}$ (e.g., via pulsed heating/evaporation), equivalent to a mass-flow pulse.

- Prediction: Excites a charge pulse

$$\boldsymbol{\Delta q = k_{mq} \frac{\Delta M}{\Delta t}},$$

and radiates an electromagnetic wave packet.

- Observables:

- Near-field: measure charge–current pulse.

- Far-field: detect radiated wave packet frequency

$$\boldsymbol{f = \frac{c\tau}{N}}$$

and compare with geometric predictions.

- Check: $[\boldsymbol{\Delta q}] = \boldsymbol{C},\ \boldsymbol{ensured\ by}\ [\boldsymbol{k_{mq}}][\boldsymbol{\Delta M/\Delta t}]$.

- Advantages: Can simultaneously test the geometric frequency relation and the bridging constant.

- Challenges: Requires $\boldsymbol{nanosecond - level\ time}$ resolution and high-speed detectors.

5.3 Table of Testable Predictions

Prediction Observable Difference from Classical Electromagnetism Falsifiability

$\boldsymbol{q = k_{mq}\dot{M}}$ (E34) Charge accumulation vs. mass flow Classical theory has no correspondence between mass flow and charge If experiment yields q=0, the model is falsified

$\boldsymbol{I \propto k_{mq}\, \omega^2 \delta M}$ (E36) AC current amplitude vs. oscillator frequency In classical EM, current depends only on external potential If $\boldsymbol{amplitude \propto \omega^2}$, supports the model

$\boldsymbol{f = c\tau/N}$ (E25) Central frequency of EM wave packet vs. geometric torsion Classical plane waves have no geometric–topological constraint If frequency deviates from geometric prediction, model is falsified

$\boldsymbol{f = m\Omega_{sp}/2}$ (E27) Resonant angular frequency vs. mode number Classical results impose no integer mode-locking If integer mode-locking is absent, the model is falsified

5.4 Error Budget (Illustrative)

- Mass measurement: microbalance/nanomechanical oscillator, sensitivity $\boldsymbol{\delta M \sim 10^{-9}\, kg}$.
- Charge measurement: electrometer, resolution $\boldsymbol{\delta q \sim 10^{-15}\, C}$.
- Frequency measurement: optical/radio-frequency locking, resolution $\boldsymbol{\delta f/f \sim 10^{-12}}$.
- Main noise sources: thermal noise, stray charges, environmental EM interference.
- Falsifiability threshold: If no charge proportional to $\boldsymbol{\dot{M}}$ is detected at the $\boldsymbol{10^{-15}\, C}$ precision level, the model's $\boldsymbol{k_{mq}}$ can be strongly constrained or ruled out.

5.5 Comparison with Maxwell and General Relativity

1. Maxwell's Electromagnetism

- Electric charge is introduced as an independent postulate.
- In the Unified Field Theory, however:

$$\boldsymbol{q = k_{mq}\,\dot{M}}$$

→ Directly testable through mass-flow experiments, which classical EM does not predict.

2. General Relativity (GR)

- GR only specifies the gravitational effects of G.
- In the Unified Field Theory, G is already locked to $\boldsymbol{\mu_g}$, but an additional mass–charge bridge is proposed.

→ This can be tested via electromagnetic experiments, independent of astronomical scales.

3. New Unified Prediction

- Geometric–frequency locking (E28):

Classical theory places no topological restrictions on EM wave frequencies.

Here, integer numbers of turns/modes are required to match.

5.6 Experimental–Reproducibility Checklist

- Define experimental system: A controllable mass-flow unit (micro-oscillator / evaporating body / loading plate).
- Synchronized measurements: Record both $\dot{M}$ and charge/current signals.
- Multiple pathways: Implement at least two of the three experimental approaches (quasi-static, resonance, pulse) to avoid systematic errors.
- Output: Determine k_{mq}, and check whether it remains constant (independent

of frequency/scale).

- Verification: Test whether the geometric frequencies (E25, E27) agree with observations.

- Error budget: Include in appendices for reproducibility.

5.7 Summary of This Part

- It is made explicit that μ_g is already locked by geometry; the only quantities requiring experimental calibration are k_{mq} and χ.

- Three complementary experimental approaches were proposed:

1. Quasi-static (direct charge measurement),

2. Resonant (frequency-dependent current response),

3. Pulse (electromagnetic radiation from transient mass unloading).

- Falsifiability condition: If charge shows no dependence on mass flow, the Unified Field Theory's mass–charge bridge is ruled out.

- The differences from traditional Maxwell/GR theories are clear and experimentally measurable.

Thus, the five parts of the paper form a closed structure:

1. Geometric scaling and locking,

2. Derivation of the mass–charge conversion,

3. Minimal action and field equations,

4. Wave solutions and frequency unification,

5. Experimental calibration and predictions.

Appendix

A.1 Table of Symbols and Dimensions

Symbol	Meaning	Dimension (SI)	Source Formula
c	Speed of light	$m \cdot s^{-1}$	Axiom
l_Ω	Rotational scaling length	m	(E1–E3)
Q	Helical divergence strength	$m^3 \cdot s^{-1}$	(E1)
μ_g	Space–mass coupling constant	$kg \cdot m^{-1}$	(E2, E3)
M	Coupled mass	kg	(E2)
G	Gravitational constant	$m^3 \cdot kg^{-1} \cdot s^{-2}$	(E3)
k_{mq}	Mass–charge conversion constant	$C \cdot s \cdot kg^{-1}$	(E8, E9)
g	Spatial flow velocity field	$m \cdot s^{-1}$	(E13)
E	Electric field $= \partial_t g$	$V \cdot m^{-1}$	(E13)
B	Magnetic field $= \nabla \times g$	T	(E13)
τ	Torsion (helical pitch density)	m^{-1}	(E25)
Ω_{sp}	Spin angular frequency	s^{-1}	(E27)

A.2 Core Formula Summary

1. Helical divergence strength:

$$Q = 4\pi c \mathrm{l}_\Omega^2$$

$$(E1)$$

2. Space–mass coupling:

$$M = \mu_g \mathrm{l}_\Omega, \quad G = \frac{4\pi c^2}{\mu_g}$$

$$(E2, E3)$$

3. Three invariants (with $u_\Omega = c$):

$$\frac{G\mu_g}{4\pi c^2} = 1, \qquad \frac{Q}{4\pi c \mathrm{l}_\Omega^2} = 1, \qquad \frac{M}{\mu_g \mathrm{l}_\Omega} = 1$$

$$(E4 - E5')$$

4. Mass–charge conversion:

$$q = k_{mq} \frac{dM}{dt}, \quad [k_{mq}] = C \cdot s/kg$$

$$(E8, E9)$$

5. Field definitions and equations:

$$E = \partial_t g, \quad B = \nabla \times g, \;\; \partial_t^2 g - \nabla \times (\nabla \times g) = J_q$$

$$(E13, E16)$$

6. Wave frequencies:

$$f_{\text{geom}} = \frac{c\tau}{N}, \qquad f_{\text{dyn}} = \frac{\mathfrak{m}\Omega_{sp}}{2\pi}, \qquad f_{\text{geom}} = f_{\text{dyn}}$$

$$(E25, E27, E28)$$

7. Experimental predictions:

$$q = k_{mq} \dot{M}, \qquad I \propto k_{mq}\, \omega^2 \delta M, \qquad f = \frac{c\tau}{N}$$

$$(E34 - E36)$$

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